

# Dynamics of Uniform Circular Motion

SS ILP Physics (Intermediate), Week 6 Class 3



# Outline

- Dynamics of Uniform Circular Motion
- Brief Discussion on Non-Uniform Circular Motion

Reference:

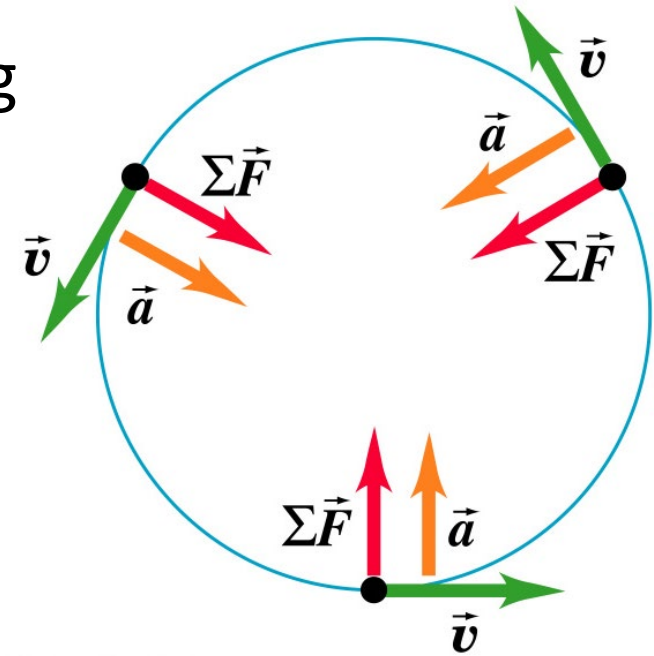
University Physics with Modern Physics

Physics for Scientists & Engineers with Modern Physics

# Dynamics of Uniform Circular Motion

- General equation (negative sign denote the force is pointing radially inward towards the center of the circle.)

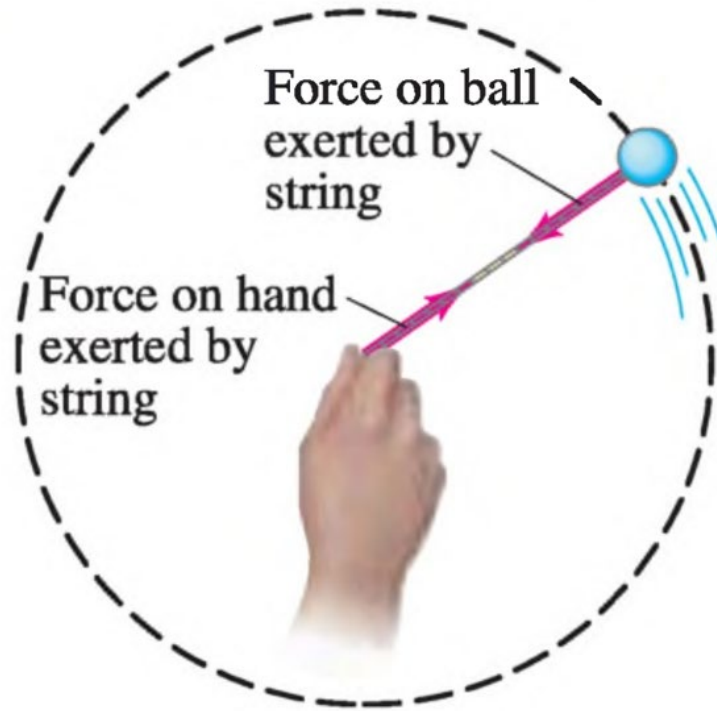
$$\sum \vec{F}_r = m\vec{a} = -mr\omega^2\hat{r} = -\frac{mv^2}{r}\hat{r}$$



- The above equation is a version of Newton's second law and means that the net force in the radial direction is given by  $-\frac{mv^2}{r}\hat{r}$ .

As we did with linear motion, the net force is never given in the free-body diagram.

# Example

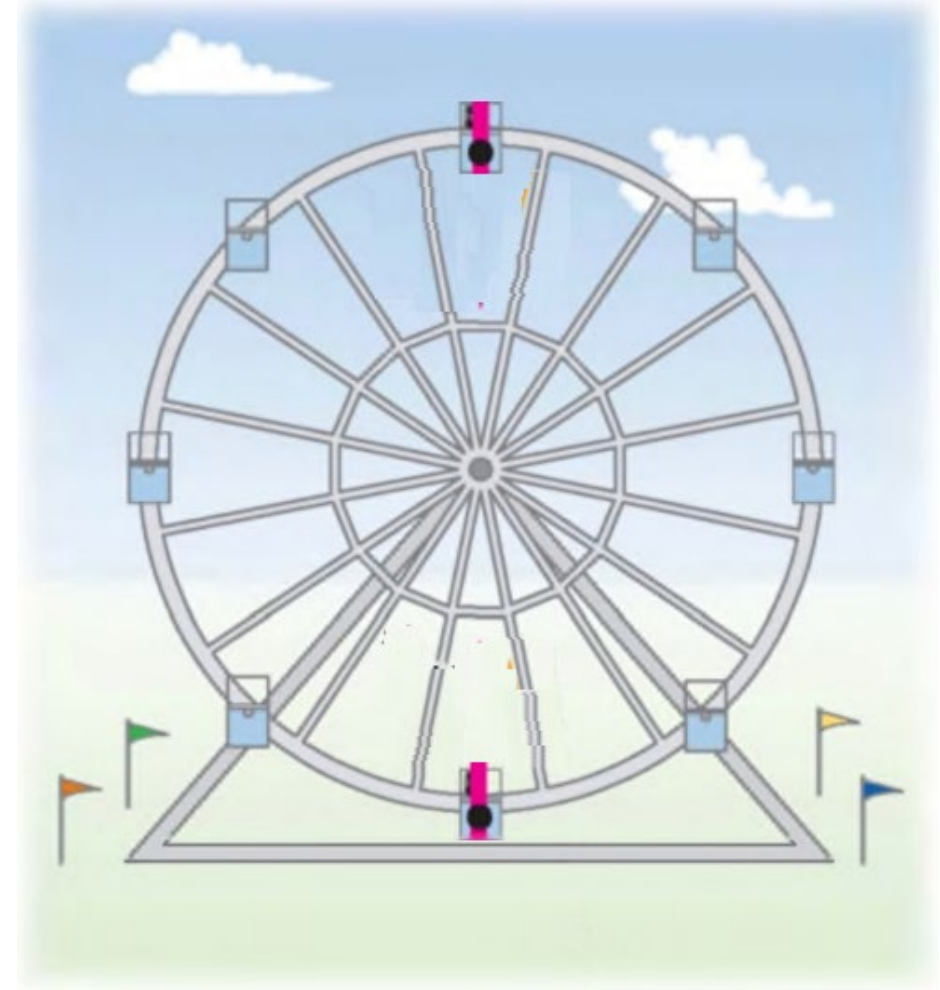


**FIGURE 5-15** Swinging a ball on the end of a string.

Physics for Scientists & Engineers with Modern Physics, Douglas C. Giancoli, Figure 5-15, p123.

# Concept Question

A rider on a Ferris wheel moves in a vertical circle of radius  $r$  at constant speed  $v$ . Is the normal force that the seat exerts on the rider at the top of the wheel (a) less than, (b) more than, or (c) the same as the force the seat exerts at the bottom of the wheel?



Physics for Scientists & Engineers with Modern Physics, Douglas C. Giancoli, Exercise F, p124.

# Thinking Aloud

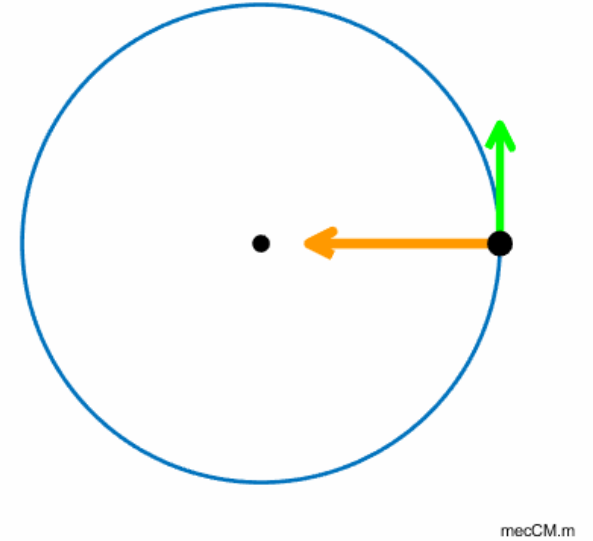
A wooden ball is attached to the rim of a spinning wheel.

The ball is held in place by a string.

When the string is cut, the ball flies in a straight line and tangent to the wheel.

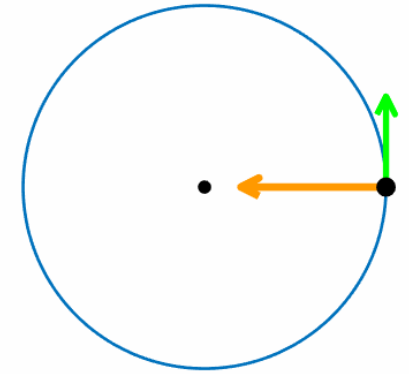
Why is this so?

Let us reason systematically.



# Thinking Aloud

- Circular motion occurs because you have tangential velocity and a centripetal force.
- The only force present is the tension in the string as a result of the spinning ball.
- If the string is cut, then there is no tension in the string and no centripetal force.
- If there is no centripetal force, then the object continues along its original tangential direction since we have tangential velocity for circular motion.



mecCM.m

Green vector is tangential velocity vector and orange vector is centripetal acceleration vector.

# Concept Question

Estimate the force a person must exert on a string attached to a 0.15-kg ball to make the ball revolve in a horizontal circle of radius 0.6 m. The ball makes 2.00 revolutions per second. Ignore the string's mass.

**FIGURE 5-17** Example 5-11.



Physics for Scientists & Engineers with Modern Physics, Douglas C. Giancoli, Example 5-11, p123.



# Newton's 2<sup>nd</sup> Law for uniform Circular Motion Dynamics

- **Step 0:**

- Draw a schematic diagram of the problem.
- Identify the coordinate system, i.e. where are the horizontal and vertical directions.
- Choose a sign convention for radial direction and check that signs for forces and acceleration are consistent.
- Identify the problems you are asked to solve.

- **Step 1:** Draw Free Body Diagram (FBD) for all physical forces

(note: centrifugal force is not a physical force and do not include  $-\frac{mv^2}{r}\hat{r}$  in the FBD).

# Newton's 2<sup>nd</sup> Law for uniform Circular Motion Dynamics

- **Step 2:** Set up Newton's 2<sup>nd</sup> Law (*aka* equation of motion)

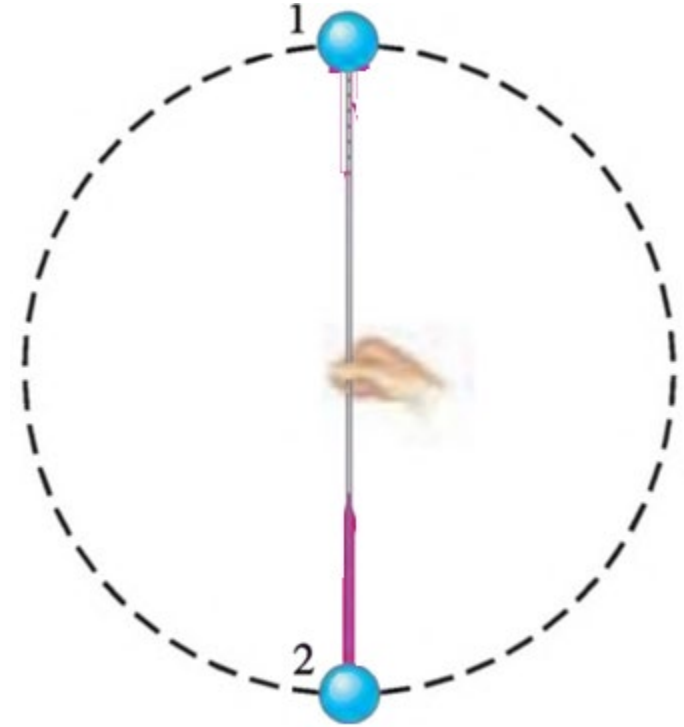
Radial direction ( $\hat{r}$ )

$$\sum F_r = ma_r$$

- **Step 3:** From the FBD in Step 1, compute  $\sum F_r$ .
- **Step 4:** Write down the net acceleration for uniform circular motion,  $a_r = -\frac{v^2}{r}$
- **Step 5:** Solve Newton's 2<sup>nd</sup> Law (Step 2) using Steps 3 and 4. Compute the problems you are asked to solve.

# Example

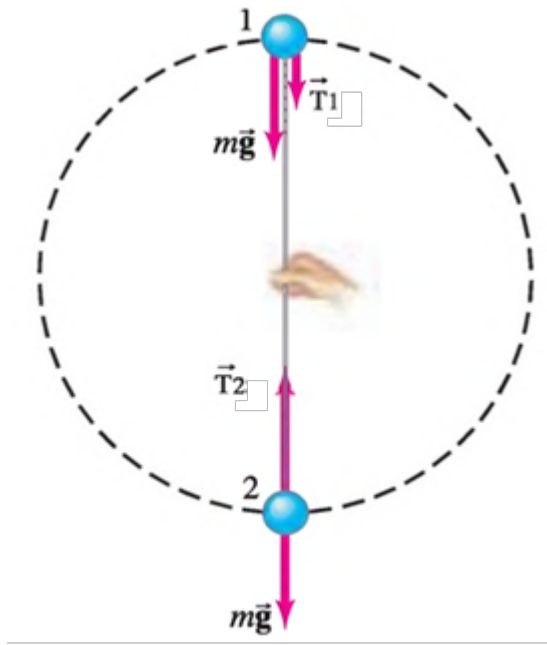
A 0.15-kg ball on the end of a 1.10 m-long cord (negligible mass) is swung in a vertical circle. (a) Determine the minimum speed the ball must have at the top of its arc so that the ball continues moving in a circle. (b) Calculate the tension in the cord at the bottom of the arc, assuming the ball is moving at twice the speed of part (a).



Physics for Scientists & Engineers with Modern Physics, Douglas C. Giancoli, Example 5-12, p124.

# Example: Solution

(a) Determine the minimum speed the ball must have at the top of its arc so that the ball continues moving in a circle.



$$T_1 + mg = ma_r = m \frac{v_1^2}{r}$$

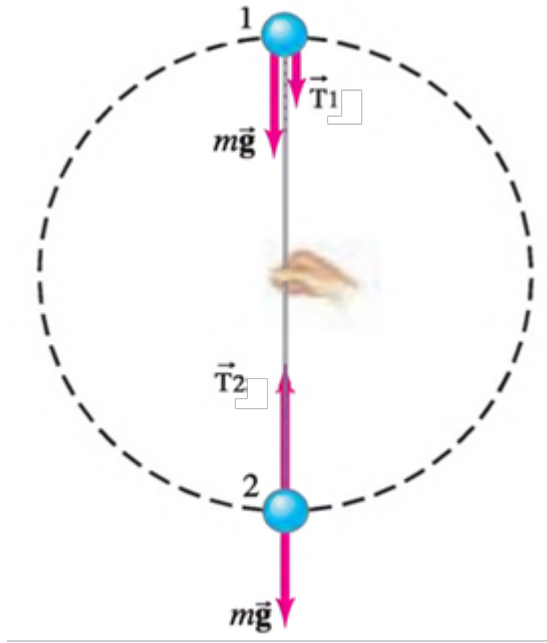
Thus, minimum speed will occur if  $T_1 = 0$ ,

$$mg = m \frac{v_{1,min}^2}{r} \rightarrow v_{1,min} = \sqrt{gr} = 3.28 \text{ m/s}$$

Physics for Scientists & Engineers with Modern Physics, Douglas C. Giancoli, Example 5-12, p124.

# Example: Solution

(b) Calculate the tension in the cord at the bottom of the arc, assuming the ball is moving at twice the speed of part (a).



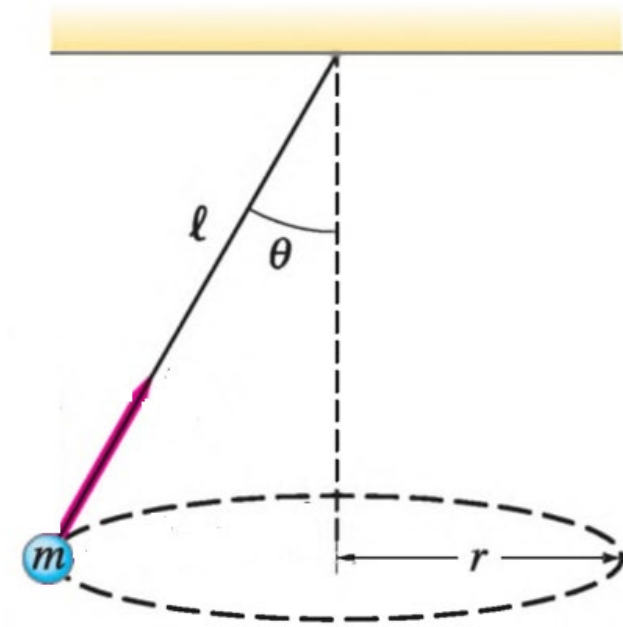
$$T_2 - mg = m \frac{v_2^2}{r}$$

$$T_2 = mg + m \frac{v_2^2}{r} = mg + m \frac{(2v_{1,min})^2}{r} = 7.35 \text{ N}$$

Physics for Scientists & Engineers with Modern Physics, Douglas C. Giancoli, Example 5-12, p124.

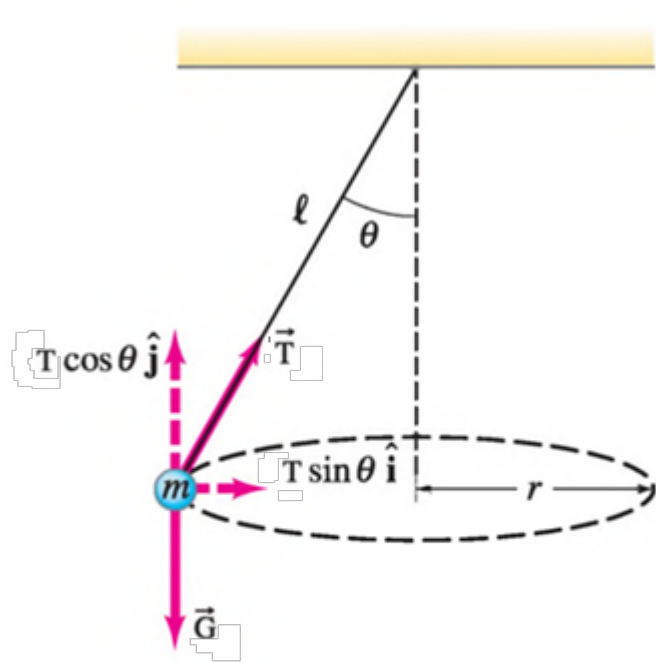
# Example

A small ball of mass  $m$ , suspended by a cord of length  $l$ , revolves in a circle of radius  $r = l \sin \theta$ , where  $\theta$  is the angle the string makes with the vertical. (a) In what direction is the acceleration of the ball, and what causes the acceleration? (b) Calculate the speed and period (time required for one revolution) of the ball in terms of  $l$ ,  $\theta$ ,  $g$ , and  $m$ .



# Example: Solution

(a) In what direction is the acceleration of the ball, and what causes the acceleration?



**FIGURE 5-20** Example 5-13.  
Conical pendulum.

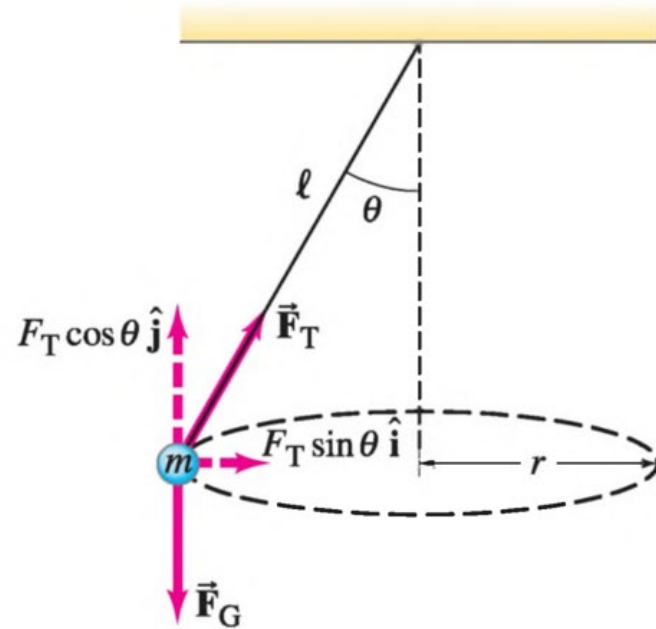
The acceleration points horizontally toward the center of the circular path.

The net force causes the acceleration.

Physics for Scientists & Engineers with Modern Physics, Douglas C. Giancoli, Example 5-13, p125.

# Example: Solution

(b) Calculate the speed and period (time required for one revolution) of the ball in terms of  $l$ ,  $\theta$ ,  $g$ , and  $m$ .



**FIGURE 5-20** Example 5-13.  
Conical pendulum.

$$\hat{r}: F_T \sin \theta = m \frac{v^2}{r}$$

$$\hat{j}: F_T \cos \theta - mg = 0 \rightarrow F_T = \frac{mg}{\cos \theta}$$

$$v = \sqrt{\frac{F_T \sin \theta r}{m}} = \sqrt{\frac{lg \sin^2 \theta}{\cos \theta}}$$

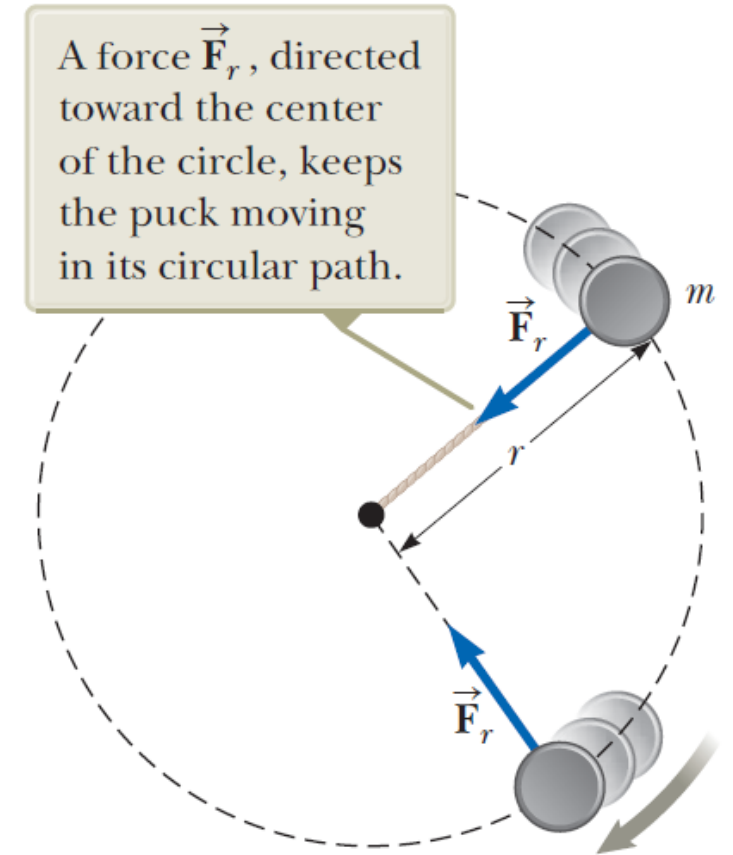
$$T = \frac{2\pi r}{v} = 2\pi \sqrt{\frac{l \cos \theta}{g}}$$

Physics for Scientists & Engineers with Modern Physics, Douglas C. Giancoli, Example 5-13, p125.



## Practice Question 1: How Fast Can it Spin?

A ball of mass 0.50 kg is attached to the end of a cord 1.50 m long. The ball is whirled in a horizontal circle as shown in the figure below. If the cord can withstand a maximum tension of 50.0 N, what is the maximum speed at which the ball can be whirled before the cord breaks? Assume the string remains horizontal during the entire motion.



An overhead view of a puck moving in a circular path on the horizontal plane.

## Practice Question 2

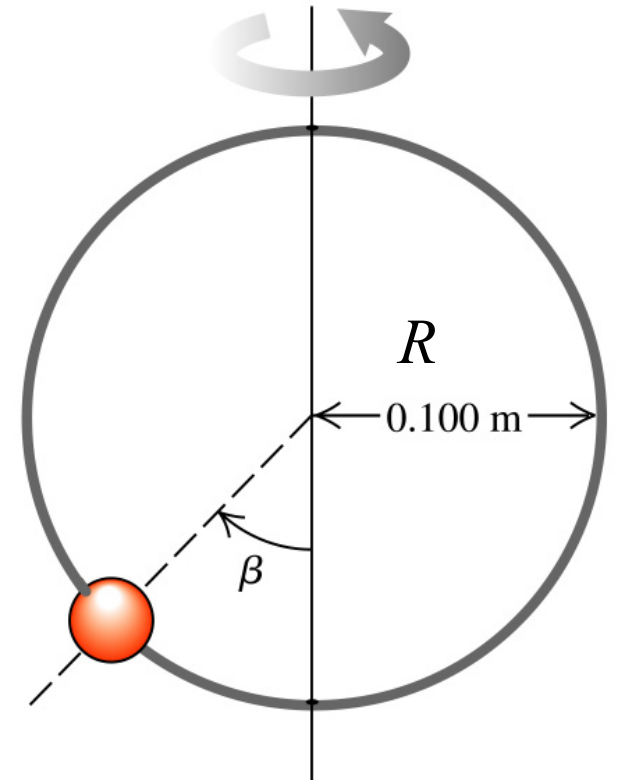
A car travels at a constant speed of 13.4 m/s on a level circular turn of radius 50.0 m. What minimum coefficient of static friction,  $\mu_s$ , between the tires and the roadway will allow the car to make the circular turn without sliding?



## Practice Question 3

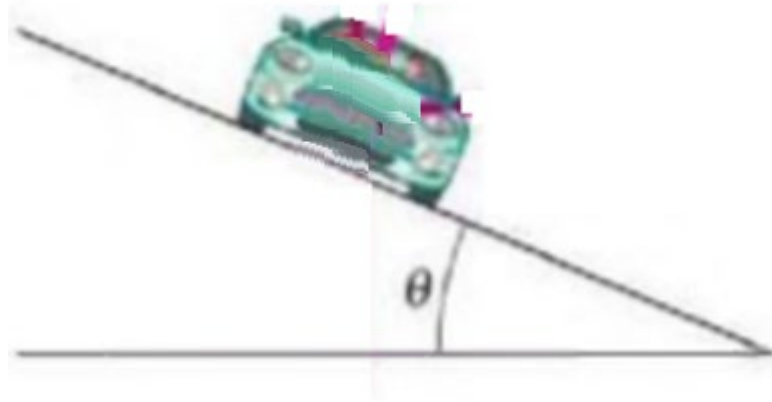
A small bead can slide without friction on a circular hoop that is in a vertical plane and has a radius of 0.100 m. The hoop rotates at a constant rate of 4.00 rev/s about a vertical diameter.

- (a) Find the angle  $\beta$  at which the bead is in vertical equilibrium.
- (b) Is it possible for the bead to “ride” at the same elevation as the center of the hoop?
- (c) What will happen if the hoop rotates at 2.00 rev/s?



## Practice Question 4

(a) For a car travelling with speed  $v$  around a curve of radius  $r$ , determine a formula for the angle at which a road should be banked so that no friction is required. (b) What is this angle for an expressway off-ramp curve of radius 50 m at a design speed of 50 km/h?



Physics for Scientists & Engineers with Modern Physics, Douglas C. Giancoli, Example 5-15, p127.